Chapter 3

TECHNICAL BACKGROUND

This chapter provides a comprehensive overview of the technologies, tools, and platforms utilized in the development and deployment of the Online Voting System. The technical foundation encompasses frontend and backend frameworks, database management systems, development tools, and deployment platforms that collectively enable the creation of a secure, scalable, and user-friendly electoral system.

Development Software Platforms

Frontend Technologies

React

React is a widely used JavaScript library for building dynamic and interactive user interfaces. It follows a component-based architecture, enabling developers to create reusable and modular UI components that enhance maintainability and scalability.

One of React's key advantages is its Virtual DOM, which efficiently updates only the necessary parts of a webpage, resulting in optimized performance and smooth user experiences. This makes React particularly effective for highly interactive applications, such as single-page applications (SPAs) and progressive web apps (PWAs).

HTML

HTML (Hypertext Markup Language) is the building block of the web, providing the foundation for structuring and displaying content on websites. It is a markup language that uses a system of tags and elements to define different parts of a webpage, such as headings, paragraphs, images, links, tables, and forms.

CSS

CSS is a fundamental technology in web development, used to define the visual presentation and layout of HTML elements. It enables developers to control colors, fonts, spacing, positioning, and animations, ensuring that web pages are both aesthetic and user-friendly. One of CSS's core advantages is its cascading nature, which allows styles to be applied hierarchically.

TypeScript

TypeScript is a superset of JavaScript that introduces static typing, enabling developers to catch errors early and write more robust and maintainable code. Developed by Microsoft, it enhances JavaScript with strong type definitions, interfaces, and modern ES6+ features, making development more predictable and scalable. With better tooling support, TypeScript improves the developer experience by providing auto completion, code navigation, and intelligent refactoring.

Backend Technologies

NestJS

NestJS is a progressive Node.js framework for building efficient, reliable, and scalable server-side applications. It uses modern JavaScript and is built with TypeScript, combining elements from Object-Oriented Programming, Functional Programming, and Functional Reactive Programming. NestJS provides a robust architecture that is heavily inspired by Angular, making it familiar for developers who have worked with Angular.

PostgreSQL

PostgreSQL is an advanced, open-source relational database management system that emphasizes extensibility and standards compliance. Known for its

robustness, feature-richness, and performance, PostgreSQL supports both relational and non-relational data types, making it highly versatile for modern applications.

Prisma ORM

Prisma is a next-generation ORM (Object-Relational Mapping) tool that provides a type-safe database access layer for modern applications. It consists of three main components: Prisma Client, Prisma Migrate, and Prisma Studio. Prisma Client is an auto-generated query builder that provides type safety and auto-completion, making database operations both efficient and error-free.

Development Environments

API Development and Communication

REST APIs

REST (Representational State Transfer) is an architectural style for designing networked applications, particularly web services. RESTful APIs use standard HTTP methods (GET, POST, PUT, DELETE) to perform operations on resources identified by URLs. This approach provides a stateless, scalable, and platform-independent way to build web services that can be consumed by various client applications.

REST APIs are designed around resources and use HTTP status codes to communicate the outcome of operations. They support multiple data formats, with JSON being the most commonly used due to its lightweight nature and widespread support. RESTful design principles promote cacheable responses, uniform interfaces, and layered system architecture, making APIs more maintainable and scalable.

WebSockets

WebSockets provide full-duplex communication channels over a single TCP connection, enabling real-time, bidirectional communication between clients and servers. Unlike traditional HTTP requests, WebSockets maintain persistent connections, allowing servers to push data to clients instantly without the need for constant polling.

This technology is essential for applications requiring real-time updates, such as voting systems with live result displays, chat applications, live notifications, and collaborative tools. WebSockets significantly reduce latency and server overhead compared to traditional polling methods, making them ideal for applications that need instant data synchronization and real-time user interactions.

Development Tools and Database Management

Visual Studio Code (VS Code)

Visual Studio Code (VS Code) is a powerful and lightweight source code editor developed by Microsoft, designed to enhance the coding experience for developers across various programming languages. It provides an intuitive and efficient environment for writing, editing, and debugging code, making it a go-to tool for both beginners and experienced developers. One of VS Code's standout features is its rich extension ecosystem, allowing users to customize and extend functionality to suit their development needs.

PGAdmin4

PGAdmin4 is a comprehensive web-based administration and development platform for PostgreSQL databases. It provides a user-friendly graphical interface for managing PostgreSQL servers, databases, and database objects. The platform offers features for database design, query execution, server monitoring, and backup management.

Postman

Postman is a comprehensive platform for API development that simplifies the process of building, testing, and managing APIs. It provides an intuitive interface for creating and executing HTTP requests, allowing developers to test REST APIs, GraphQL endpoints, and WebSocket connections efficiently.

Tools and Technologies Used

Version Control and Collaboration

GitHub

GitHub is a widely used platform that enables developers, teams, and organizations to collaborate on software development projects efficiently. It provides a web-based Git version control system, allowing users to manage, store, and track changes in their code repositories. GitHub simplifies teamwork by offering tools such as pull requests, code reviews, and issue tracking, making it easier to propose, discuss, and implement updates seamlessly.

Git

Git is a distributed version control system designed to handle everything from small to very large projects with speed and efficiency. It allows developers to track changes in source code during software development, enabling multiple contributors to work on a project simultaneously without overwriting each other's work. Git manages code history through commits, branches, and merges, giving developers the flexibility to experiment and revert changes when necessary.

Hardware Requirements

Development Environment Specifications

- Laptop: ROG STRIX G15 (2022) G513RC-HN016W – A high-performance gaming laptop used for development.
- Processor: AMD Ryzen 7 6800H – Ensures fast and efficient execution of applications and development tools.

- RAM: 16GB DDR5-4800MHz – Provides smooth multitasking and fast data processing for development workflows.
- Storage: 512GB SSD + 1TB NVMe SSD – Offers high-speed storage for quick application loading, build processes, and data access.
- Operating System: Windows 11 Pro – Provides a stable and optimized environment for software development with support for modern development tools.
- Graphics: NVIDIA GeForce RTX 3050 – Supports graphical processing for UI development, testing, and other tasks requiring GPU acceleration.